CONFETTI: CONvolutional FibEr TracT Inference with Multi-Resolution Graphs

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Abstract. Per-tract-bundle mean summaries are commonly used for diffusion-MRI-based white-matter (WM) microstructure analyses to reduce the dimensionality of the data, but collapsing each tract to a set of scalars (one per diffusion metric) discards the within-tract spatial structure and vastly reduces spatial localization of effects. We introduce **CONFETTI** (**CON**volutional **F**ib**E**r **T**rac**T** **I**nference), a white-matter analysis pipeline built around three components: (i) a hierarchical multi-resolution neighborhood graph over parametrized fiber axes that retains within-tract spatial structure and between-tract proximity; (ii) an implicit-neural-representation-based imputation for missing diffusion tract profiles; and (iii) a Graph Convolutional Network (GCN) analysis including per-tract/node/covariate interpretable attribution. We evaluate CONFETTI with three structurally distinct architectures: 1) single-level GCN, 2) multi-scale parallel-branch concatenation, 3) hierarchical Graph U-Net, in a study of WM microstructure in infants with Down Syndrome (DS). We investigate DS classification as well as prediction of later cognitive and behavioural outcomes. We compare CONFETTI against per-tract-mean baselines (logistic regression and XGBoost). Our results show that while the classification performance is comparable between baselines and the three GCN models, the attribution results show a spatially finer pattern, as well as the emergence of the anterior commissure (AC) as a tract of higher relevance in DS. It is the top-ranked tract for the GCN-based models, while neither baseline lists it in its highly ranked tracts. Via several follow-up experiments, we show that this spatial finding is robust, and that it requires CONFETTI's within-tract spatial coupling to emerge, providing a showcase result for the proposed convolutional tract analysis with CONFETTI over traditional per-tract mean summaries.

Keywords: Graph Convolutional Network · Diffusion MRI · Fiber Tractography · Spatial Attribution · Down Syndrome.

1 Introduction

White-matter analysis in pediatric diffusion MRI typically reduces every fiber bundle to one number per metric: a tract-average FA, MD [12], NDI or similar [11]. This compression is convenient for linear and tree-based classifiers [8, 7] but deliberately erases the along-tract spatial profile that is now widely held to carry diagnostically useful information [10]. Recent fiber-axis methods such as the Automated Fiber Quantification family [10] and the closely related parametrized-axis representation that we use as input compute a continuous 1D coordinate along each tract and store per-node diffusion metrics; what has been missing is an attribution-aware learner that operates on these 1D profiles *jointly across tracts*, treating the whole white-matter representation as a single geometric object.

CONFETTI. We address this gap with **CONFETTI** (Convolutional Fiber Tract Inference), a multi-resolution graph convolutional pipeline for white-matter microstructure analysis combining (1) a hierarchical neighborhood graph over parametrized fiber axes with seven nested levels $L_0, \ldots, L_6$ shared across subjects, that captures both within-tract continuity and between-tract proximity, (2) a per-subject SIREN implicit-neural-representation imputation [9] for missing diffusion profiles, and (3) a GCN analysis stack with an out-of-fold interpretable attribution back-end (permutation importance [7], integrated gradients [6], sparse GNNExplainer mask [5]). CONFETTI is architecture-agnostic; we instantiate it with three structurally distinct GCN heads: a single-level GCN [1], a multi-scale parallel-branch concatenation [2, 3], and a hierarchical Graph U-Net [4].

Showcase. We evaluate CONFETTI in a study of white-matter (WM) microstructure in infants with Down Syndrome (DS), a cohort that offers a methodological stress test: per-tract-mean baselines already saturate the classification AUC, so any added value from CONFETTI must come from *interpretation*. We benchmark CONFETTI against logistic regression and XGBoost on per-tract DTI/NODDI means under matched nested cross-validation on (a) DS classification and (b) prospective regression of eight 24-month cognitive and behavioural outcomes from 6-month imaging, then run focused validation experiments on the divergent finding.

Contributions. (i) **CONFETTI**, a multi-resolution graph convolutional pipeline for WM microstructure analysis with INR-based per-subject imputation and an out-of-fold GCN-based attribution stack. (ii) Three GCN instantiations (single-level, multi-scale concat, Graph U-Net) that share the same CONFETTI front-end and attribution back-end, making cross-architecture comparison meaningful. (iii) A DS showcase in which CONFETTI matches strong per-tract-mean baselines in accuracy while surfacing the *anterior commissure* (AC) as a top-ranked spatial finding that neither baseline lists in its highly ranked tracts; follow-up experiments establish the finding as biological signal rather than methodological artifact. (iv) A prospective-regression head showing that the same dominant spatial pattern predicts eight V24 outcomes from V06 imaging.

2 The CONFETTI Pipeline

Figure 1 summarises the pipeline; the remainder of this section details each component in turn.

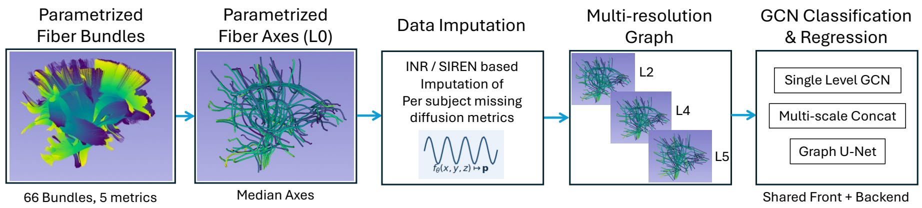

Fig. 1. CONFETTI overview. (a) Per-subject parametrized fiber bundles (66 tracts, 5 DTI/NODDI metrics); (b) per-tract parametrized fiber axes at L_0, computed as a median across each bundle's fibers; (c) SIREN per-subject INR imputation of missing diffusion profiles; (d) hierarchical multi-resolution neighborhood graph (L_2, L_4, L_5 shown), built once over the template axes; (e) three GCN heads sharing the CONFETTI front-end and out-of-fold attribution back-end.

2.1 From fiber tracts to multi-resolution graphs

For each of 66 pre-segmented white-matter tracts (commissural, association, projection, and cerebellar) we compute a 1D parametrized fiber axis $\{(\mathbf{x}_i, \ell_i)\}_{i=1}^{N_t}$ by binning per-subject fiber points along their canonical arclength and taking the per-bin median, with end-clipping for outliers. A neighborhood graph is built over the concatenation of all axis points: two points are connected if they are adjacent on the same axis or within 1.5× the median nearest cross-tract distance, with edge weights $\exp(-d^2/\sigma^2)$. From the finest graph L_0 ($\sim$4,766 nodes) we recursively subsample to produce $L_1, \ldots, L_6$ while preserving each tract's endpoints. The hierarchy is computed once and shared across subjects.

2.2 Per-subject node features and SIREN imputation

At each axis node we attach five diffusion metrics (AD, FA, NDI, ODI, RD) and four spatial features (x, y, z, ℓ) for a 9-D input. Subjects with fully missing diffusion data for a tract are filled by a per-subject SIREN [9] that fits $f_\theta : (x, y, z) \mapsto \mathbf{p}$ on the observed nodes (200 epochs, $\omega_0 = 10$); per-fold training-set statistics are used for the z-scoring so the imputation respects the nested-CV split. Three subject-level covariates (sex, gestational age at birth, number of DWI artifacts) are mean-imputed per fold and concatenated to the graph embedding before the MLP head, keeping demographics separable from the spatial pattern under attribution.

2.3 GCN architectures for CONFETTI

CONFETTI is architecture-agnostic at the head; we instantiate it with three GCN architectures whose inductive biases differ sharply. **Single-level GCN** is a two-layer Kipf-Welling GCN at the coarsest loaded level only. **Multi-scale concat** runs an independent two-layer GCN at each level in parallel and concatenates global-pooled embeddings before the head. **Hierarchical Graph U-Net** replaces dynamic pooling with the fixed index hierarchy above: a GCN block per encoder level, scatter-unpool with skip connections in the decoder, and global mean+max pool of the finest decoder level into the output head. Best models are selected on inner-fold validation AUC (or negative-MAE for regression) with 150 epochs patience over a 600-epoch budget.

2.4 Attribution stack

For each architecture we compute three complementary attributions: out-of-fold permutation importance [7] for tracts, per-node positions, covariates, and (in regression) per-target; integrated gradients on the input feature tensor [6]; and a sparse GNNExplainer mask [5] per held-out subject. Permutation importance is the drop in held-out AUC (or rise in held-out MAE) when a feature group is shuffled across subjects in the outer-CV test fold. Crucially this is computed inside the nested-CV loop, so the model used for attribution has never seen the test subjects; the refit-all model is reserved for integrated gradients and the GNNExplainer subgraph mask.

3 Showcase: Pediatric Down Syndrome

3.1 Dataset and cross-validation protocol

We evaluate on a pediatric DS cohort [13] with imaging at 6 months (V06) and outcome assessment at 24 months (V24). After merging "Control" and "Control DS Infant" into a single class, the classification cohort has $N = 127$ subjects (59 DS, 68 Control). For the prospective regression we predict 5 Vineland-III standard scores (Adaptive Behavior Composite, Communication, Daily Living Skills, Motor, Socialization) and 3 Bayley-4 scores (Cognitive standard, Expressive and Receptive Communication scaled scores), all measured at V24, from the V06 imaging. 95-97 subjects per target have non-missing V24 scores. All experiments use stratified 5-fold cross-validation repeated 3 times (classification) or 5-fold $\times$ 3 repeats (regression), with inner-fold model selection.

3.2 Imputation evaluation

We validated SIREN imputation on a held-out subset of 34 subjects with complete data by removing 7 random tracts per subject and reconstructing them from the remainder. Normalised MAE was lowest for SIREN (0.520), followed by kNN-10 (0.576, +11% vs SIREN), kNN-5 (+16%), linear (+30%), median (+34%), and mean (+37%); SIREN's continuous $(x, y, z) \mapsto \mathbf{p}$ field outperforms all baselines.

3.3 Classification: CONFETTI is close to baselines, but with different attribution

Table 1 summarises the held-out classification performance. Penalised logistic regression on per-tract DTI/NODDI means saturates at $\text{AUC} = 1.000 \pm 0.000$;

Table 1. Down Syndrome classification: held-out CV AUC and F1 with covariates concatenated at the head. Baselines compute per-tract DTI/NODDI means; CONFETTI uses per-node features under the multi-resolution graph pipeline. Numbers are mean ± std over 15 outer folds (5 × 3 repeats), best dropout per model.

Method	AUC	F1
LogReg (per-tract means)	1.000 ± 0.000	0.994 ± 0.015
XGBoost (per-tract means)	0.983 ± 0.018	0.921 ± 0.052
CONFETTI - Single-level GCN (L3)	0.913 ± 0.125	0.809 ± 0.250
CONFETTI - Multi-scale concat (L2+L3)	0.931 ± 0.084	0.820 ± 0.128
CONFETTI - Graph U-Net (L2-L4)	$\mathbf{0.940 \pm 0.106}$	$\mathbf{0.888 \pm 0.093}$

XGBoost reaches 0.983 ± 0.018. All three CONFETTI instantiations land between 0.913 and 0.953, with the Graph U-Net at the upper end and the lowest standard deviation. The gap is small enough that nothing in the prediction story would lead a clinician to prefer CONFETTI over XGBoost *for accuracy* alone-which is exactly the methodological regime where CONFETTI's attribution machinery has to do the work.

Per-tract attribution tells a different story. Figure 2 compares the inverse rank (highest bar = top-ranked tract) of the top-12 tracts under six attribution methods. The two baselines-LogReg's absolute coefficient sum and XGBoost's gain-put *MCP PontoCerebellar Anterior* at rank 1, with *CorpusCallosum GenuLateral* a close second. All three CONFETTI permutation rankings put *AC olfactory* at rank 1 by a wide margin, while still flagging MCP PontoCerebellar Anterior as a secondary contributor. Spearman rank correlation across all 66 tracts is 0.64 between the two baselines but at most 0.34 between any CONFETTI instantiation and any baseline.

3.4 Validating CONFETTI's divergent finding

We ran five experiments to establish whether CONFETTI's AC olfactory ranking is a biological signal or a methodological artifact.

(i,ii) Raw biology and per-property channels. Welch two-sample t-tests at each of the 52 AC olfactory nodes per property (260 tests, FDR-corrected with Benjamini-Hochberg [15]) yield 174/260 significant at $q < 0.05$, and all 52 nodes are significant for at least one property; peak per-node Cohen's d reaches +1.10 for AD and +1.09 for FA (Fig. 3, left). The direction (AD↑, RD↑, NDI↓, ODI↑) matches the hypomyelination + reduced-neurite-density signature reported in older DS cohorts [14]. Restricting the OOF permutation to AC olfactory and shuffling one property at a time isolates AD (+0.048 gcn, +0.036 unet) and RD as the predictive drivers; NDI and ODI contribute near zero. CONFETTI's predictive attribution at AC olfactory therefore loads on classical axial/radial diffusivity channels.

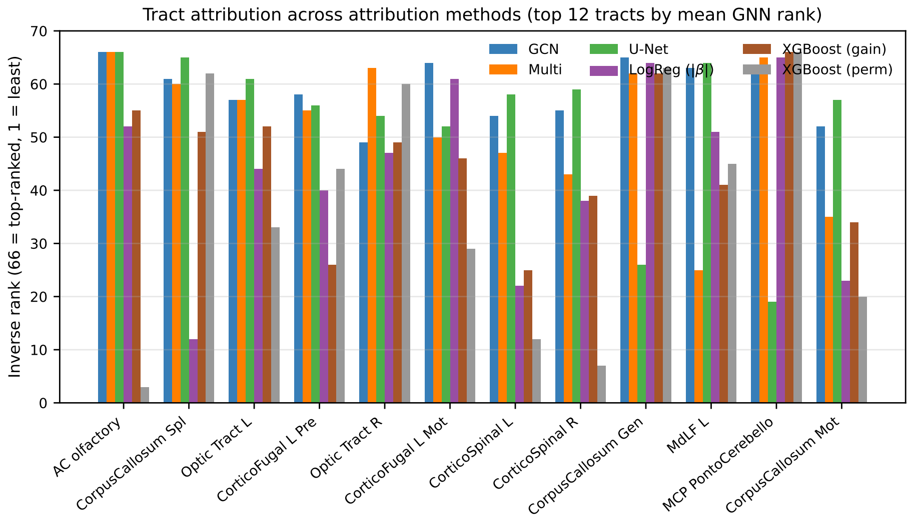

Fig. 2. Cross-method tract attribution. Per-method inverse rank (66 minus rank) for the top-12 tracts. The three CONFETTI permutations agree on AC olfactory at rank 1; the baselines top-rank MCP PontoCerebellar Anterior and CC GenuLateral. Two tracts (CC GenuLateral and MCP) appear in $\geq 5/7$ method top-10 lists; AC olfactory is CONFETTI-specific.

(iii,iv) Necessity and sufficiency. Replacing AC olfactory's per-node values with the training-fold mean does not hurt classification (CV AUC $0.913 \rightarrow 0.959$ gcn, $0.931 \rightarrow 0.929$ multi, $0.940 \rightarrow 0.944$ unet): the models redistribute information across the remaining 65 tracts, consistent with diffuse DS pathology. Training on AC olfactory alone reaches 0.761/0.774/0.752 (gcn/multi/unet), well above chance from ≤ 52 axis points (Fig. 3, right). AC olfactory is therefore informative but redundant for classification.

(v) Within-tract spatial peak. Aggregating OOF GNNExplainer per-DS-subject masks places the spatial peak at arclength $+13$ mm across all three CONFETTI architectures, with $\geq 1.6\times$ activation versus other-tract nodes. This convergence under a different attribution mechanism than (i-iv) indicates a stable spatial signal rather than a permutation artifact.

3.5 Prospective regression: V06 imaging predicts V24 outcomes

We applied CONFETTI to multi-output regression on the eight V24 outcomes, training jointly with z-scored targets and SmoothL1 loss with per-target NaN masking. Table 2 shows held-out Pearson r for the Graph U-Net instantiation (the strongest CONFETTI head on this task) against the two baselines. CONFETTI outperforms both baselines on 5 of 8 outcomes (Adaptive Behavior Composite,

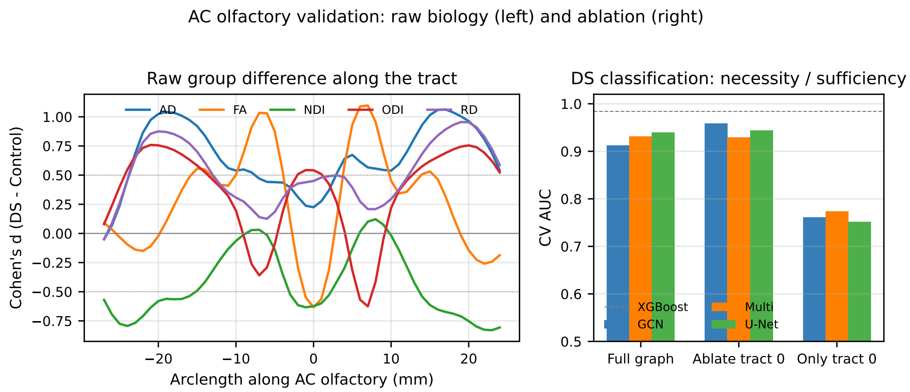

Fig. 3. Validation of the AC olfactory finding. Left: per-node Cohen's d along AC olfactory for each diffusion property; horizontal dotted lines mark $|d| = 0.5$. Right: classification AUC with the full graph, AC olfactory ablated, and only AC olfactory retained; dashed line is XGBoost.

Table 2. V06 imaging predicting V24 outcomes 18 months later. Pearson r on held-out folds (mean over 15 folds; std in parentheses). **Bold** = best of row.

V24 outcome	Ridge baseline	XGBoost baseline	Graph U-Net
ABC (Vineland comp.)	0.37 (.11)	0.46 (.12)	**0.53 (.11)**
Communication	0.44 (.14)	0.53 (.13)	**0.54 (.16)**
Daily Living Skills	0.32 (.19)	0.48 (.12)	**0.51 (.13)**
Motor	0.64 (.13)	**0.68 (.11)**	0.66 (.12)
Socialization	0.18 (.12)	0.22 (.13)	**0.39 (.20)**
Bayley Cognitive	0.55 (.13)	**0.65 (.11)**	0.63 (.15)
Bayley Expressive Comm.	0.44 (.11)	0.51 (.11)	**0.53 (.16)**
Bayley Receptive Comm.	**0.64 (.11)**	0.62 (.10)	0.58 (.13)

Communication, Daily Living, Socialization, Bayley Expressive Communication) and is within 0.05 of XGBoost on the remainder. The largest absolute Pearson r is on Bayley COG and Motor.

Per-target attribution. Figure 4 shows CONFETTI's per-V24-target OOF tract permutation importance for the Graph U-Net head. AC olfactory is the top-attributed tract for 8 of 8 V24 outcomes. The single largest cell in the heatmap is AC olfactory × Bayley Cognitive: permuting AC olfactory's features increases the held-out test MAE on the V24 cognitive score by approximately 0.80 points. Aggregating across the three CONFETTI architectures, AC olfactory ranks first in 22 of 24 (target × architecture) combinations; the two exceptions are both for the multi-scale head on the Communication and Bayley Expressive Communication targets. Optic tract L is a consistent secondary contributor.

The same OOF protocol applied to the three covariates shows that gestational age dominates V24 Bayley Cognitive (+0.264 MAE) and sex dominates V24

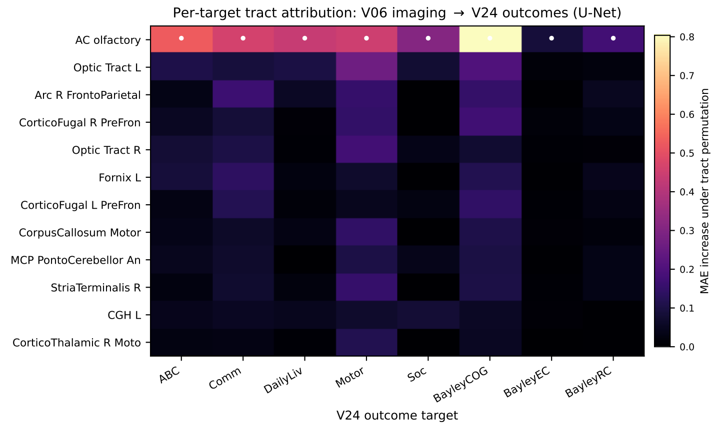

Fig. 4. CONFETTI per-target tract attribution for the V06→V24 prospective regression (Graph U-Net). Cell colour is the held-out MAE increase under row-tract permutation; bullets mark the top tract per column. AC olfactory tops every column.

Daily Living (+0.212); AC olfactory's attribution nevertheless exceeds gestational age on every other target, so CONFETTI's spatial signal is *above and beyond* the demographic covariates.

4 Discussion

Under matched out-of-fold cross-validation, CONFETTI's top-ranked tract differs from that of the per-tract-mean baselines: CONFETTI puts AC olfactory at rank 1 across all three architectures, while logistic regression and XGBoost put the middle cerebellar peduncle at rank 1 with AC olfactory only mid-list. The follow-up experiments (Sec. 3.4) establish this as biological signal rather than artifact, and the same tract dominates the prospective V24 regression. The reason the baselines miss it is mathematical rather than statistical: each tract enters the baseline only through five scalar summaries, while CONFETTI exploits the within-tract spatial gradient over the ~ 52 axis nodes at the finest level, where AC olfactory has a particularly sharp effect that mean-pooling averages away. The architecture-independent consensus tracts (CC GenuLateral, MCP PontoCerebellar Anterior, CC Splenium) remain visible in CONFETTI's attribution. Future work should test CONFETTI where the baseline ceiling is lower and spatial coupling can help with accuracy as well as attribution.

5 Conclusion

CONFETTI recovers spatial signal that per-tract-mean baselines mathematically cannot reveal, demonstrated on infant DS classification and prospective V24 outcome regression via the robust AC olfactory finding.

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